

# Satpuda Core ERP Step-by-Step Implementation Plan

Detailed implementation roadmap to transform the current app into ERP with calculation updates, database updates, UI updates, Sales Return, Purchase Return, GST reports, profit, ledger-based account clear, inventory, and exports. Documentation only; no code changes included.

## 0. First Safety Setup

- Back up full project folder and veterinary.db before any migration.
- Create a fresh test database for ERP work; do not test on live data.
- Keep current app working while new ERP tables are added beside old tables.
- After every phase, run Sales, Purchase, Inventory, Sales History, Purchase History, and exports once.

## 1. Add Migration System

- Add schema\_versions table: version, name, applied\_at, notes.
- Move all future ALTER/CREATE statements into repeatable migration functions.
- Each migration must check if table/column exists before creating it.
- Done when app can start twice without duplicate column errors.

## 2. Add Decimal Money Service

- Create core/money.py with Decimal parsing, money rounding, rupee round-off, and currency formatting.
- Use ROUND\_HALF\_UP: `money(x)=Decimal(x).quantize(0.01)`.
- All new calculation services must use Decimal, not float.
- Tests: 2.675 -> 2.68, 100.50 rounds to 101, 100.49 rounds to 100.

## 3. Add GST and Discount Calculator

- Create core/tax\_calculator.py.
- Purchase formula: `gross_taxable=qty*rate_ex_gst`; discount; taxable; GST; line\_total.
- Sales formula for MRP inclusive: `net_inclusive` after discounts; `taxable=net*100/(100+gst)`; `GST=net-taxable`.
- Allocate bill discount line-wise before GST for filing accuracy.
- Split GST into CGST/SGST or IGST using supply\_type.

## 4. Add ERP Database Tables

- Add medicine\_master for name/type/manufacturer/content/HSN/default GST/schedule.
- Add medicine\_batches with supplier\_id, purchase\_id, purchase\_item\_id, batch, expiry, unit\_factor, stock\_qty\_available, cost\_rate\_ex\_gst, MRP, location.
- Add party\_ledger, payments, payment\_allocations, stock\_ledger, invoice\_sequences, audit\_log.
- Add sale\_returns/sale\_return\_items and purchase\_returns/purchase\_return\_items.

## 5. Update Sales Tables

- Add sales header fields: taxable\_total, cgst\_total, sgst\_total, igst\_total, gst\_total, item\_discount\_total, bill\_discount\_type/input/amount, round\_off, grand\_total, invoice\_status, payment\_status, return\_total, void fields.
- Add sales\_items fields: batch snapshot, expiry snapshot, HSN, unit\_factor, item discount input/amount, bill\_discount\_allocated, taxable\_value, GST split, line\_total, cost\_rate\_ex\_gst\_snapshot, returned\_qty.
- Done when a sale item stores all values required for GST, profit, and return without reading current medicine values.

## 6. Update Purchase Tables

- Add purchases fields: supplier\_bill\_date, rate\_tax\_mode, taxable\_total, input GST split totals, discount totals, round\_off, grand\_total, invoice\_status, payment\_status, return\_total, void fields.
- Add purchase\_items fields: created\_batch\_id, gross\_taxable\_amount, discount input/amount, bill\_discount\_allocated, taxable\_value, GST split, line\_total, unit\_factor, stock\_qty\_added, returned\_qty.
- Done when Purchase GST report can come only from purchase\_items.

## 7. Build Ledger Service

- Create core/ledger\_service.py.
- Customer signs: sale +, payment -, sales return -, refund +. Balance >0 due, 0 clear, <0 credit.
- Supplier signs: purchase +, payment -, purchase return -. Balance >0 payable, 0 clear, <0 advance/credit.
- Add FIFO allocation: payment clears oldest open invoices; extra payment becomes credit/advance.
- History should show invoice\_status and party\_balance separately.

## 8. Build Stock Service

- Create core/stock\_service.py.
- Post stock in for purchases and sales returns. Post stock out for sales and purchase returns.
- Every stock change writes stock\_ledger with source\_type and source\_id.
- Batch stock\_qty\_available must equal opening + ins - outs.
- Do not directly edit stock without stock adjustment entry and reason.

## 9. Rebuild Purchase Save Flow

- Purchase should calculate using tax\_calculator, then save header, items, batch rows, stock ledger, supplier ledger, payments, allocations in one transaction.
- Free quantity increases stock but not taxable value unless supplier bill values it.
- Supplier credit/advance must be applied explicitly and recorded.
- Done when mixed GST purchase with discount, free qty, and partial payment saves correct stock/GST/supplier balance.

## 10. Rebuild Purchase UI

- Header: supplier, GSTIN, bill no, supplier bill date, rate tax mode, payment terms.
- Item grid: medicine, batch, expiry, qty, free qty, unit factor, rate, discount, taxable, GST, line total, MRP, location.
- Summary: taxable total, input GST, discount, round-off, grand total, supplier credit used, paid, payable, supplier balance after.

- Update web purchase helper JSON to include the same tax and discount fields.

## 11. Rebuild Sales Save Flow

- Sales should select from medicine\_batches with available stock.
- Calculate GST-inclusive MRP lines and store taxable/GST/cost snapshots.
- Post stock out, customer ledger +grand\_total, payment rows, and allocations.
- Customer credit must reduce payable only through recorded credit\_used allocation.
- Done when sale reprint, GST report, profit report, and customer due do not depend on current medicine values.

## 12. Rebuild Sales UI and Bill Print

- Item grid: medicine, batch, expiry, stock, qty, MRP, discount, taxable, GST, total.
- Summary: taxable, GST, discount, round-off, grand total, credit used, cash, online, due, balance after.
- Bill print must read stored sale item tax fields, not current medicines.gst\_percent.
- Keep scheduled medicine doctor validation.

## 13. Rebuild Purchase History

- Columns: bill, date, supplier, taxable, input GST, discount, round-off, grand total, paid, returns, invoice due, supplier balance, status.
- Actions: view, print, payment, purchase return, void, correction, export.
- Show detail drawer: items, GST lines, payments, ledger, stock movements, audit.
- Do not hard delete posted purchases; void with reversal entries.

## 14. Rebuild Sales History

- Columns: bill, date, customer, taxable, GST, discount, round-off, grand total, paid, returns, invoice due, customer balance, status.
- Actions: view, print, receive payment, sales return, void, correction, export.
- Profit uses sale taxable value - cost snapshot and reverses returns.
- Old cleared invoices remain cleared even after a new due invoice.

## 15. Add Sales Return

- Open return from original sale only. Show sold qty, returned qty, returnable qty.
- $\text{return\_qty} \leq \text{sold\_qty} - \text{already\_returned\_qty}$ .
- Use original sales\_items taxable/GST/cost snapshots.
- Post sale\_return, sale\_return\_items, stock in, customer ledger -return\_total, refund payment if any.
- Update sale status to PARTIAL\_RETURN or RETURNED.

## 16. Add Purchase Return

- Open return from original purchase only. Show purchased qty, free qty, returned qty, available stock.

- Validate stock is available before return.
- Use original purchase\_items taxable/input GST snapshots.
- Post purchase\_return, purchase\_return\_items, stock out, supplier ledger -return\_total.
- Settlement modes: debit note, supplier credit, cash refund.

## 17. Add GST Report Center

- Reports: Sales GST, Purchase GST, Sales Return GST, Purchase Return GST, Net GST, HSN Summary.
- Use stored taxable and GST fields. Never recalculate GST from medicine master.
- Net GST = output GST - input GST after return adjustments.
- Export CSV, Excel, and printable HTML/PDF with date range and totals.

## 18. Add Profit and Inventory Reports

- Profit = sale taxable value - cost\_rate\_ex\_gst\_snapshot\*qty, less sales return reversal.
- Stock valuation = batch available qty \* cost\_rate\_ex\_gst and optional MRP value.
- Reports: item profit, monthly profit, doctor-wise profit, stock movement, stock valuation, expiry, reorder.

## 19. Add Export Center

- Create one Export Center page grouped by Sales, Purchase, GST, Inventory, Accounts, Returns, Audit.
- Reuse core/export\_manager but feed it report\_service rows.
- Every export should include filters, generated time, totals, and store profile.

## 20. Add ERP Dashboard and Navigation

- Group navigation: Dashboard, Sales, Purchase, Inventory, Accounts, GST, Reports, Masters, Admin.
- Dashboard cards: today sales, collection, purchases, customer due, supplier payable, low stock, expiry, GST estimate, pending returns.
- Add quick actions: New Sale, New Purchase, Sales Return, Purchase Return, GST Report, Export Center.

## 21. Add Void, Correction, and Audit

- Replace destructive delete for posted documents with void.
- Void posts reversing stock and ledger entries and requires reason.
- Correction should void/repost or create correction link to original.
- Every void/correction writes audit\_log with user, time, old/new data.

## 22. Add Roles and Permissions

- Roles: Admin, Billing, Purchase, Inventory, Accountant, Viewer.
- Restrict void, correction, settings, delete all, export all, backup restore.
- Require admin PIN/password for risky actions and log them.

## 23. Final Test Matrix

- Purchase with mixed GST, bill discount, free qty, partial payment.
- Sale with mixed GST, customer credit, partial payment.
- Sales return partial: stock increases, GST/profit/customer balance reverse.
- Purchase return partial: stock decreases, input GST/supplier balance reverse.
- Void sale and void purchase create reversal entries.
- All GST/profit/stock/ledger exports match manual totals.

## Minimum Release Order

- v1 Foundation: steps 1-8
- v2 Purchase: steps 9-10
- v3 Sales: steps 11-12
- v4 Histories: steps 13-14
- v5 Returns: steps 15-16
- v6 Reports: steps 17-19
- v7 ERP polish: steps 20-23

## Final Rule

Do not implement Sales Return and Purchase Return before item GST snapshots, cost snapshots, stock ledger, and party ledger are working. Returns are only reliable when original sale/purchase lines already store the exact taxable, GST, discount, batch, supplier, and cost values.